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3D TCM with damping

We study the following problem

$$\begin{aligned} &\mathbf{u}_t + (\mathbf{u} \cdot \nabla) \mathbf{u} - \Delta \mathbf{u} + |\mathbf{u}|^{\alpha-1} \mathbf{u} + \operatorname{div}(\mathbf{v} \otimes \mathbf{v}) = \nabla p, \\ &\operatorname{div} \mathbf{u} = 0, \\ &\mathbf{v}_t + (\mathbf{u} \cdot \nabla) \mathbf{v} - \Delta \mathbf{v} + |\mathbf{v}|^{\beta-1} \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{u} = \nabla \theta, \\ &\theta_t + (\mathbf{u} \cdot \nabla) \theta - \Delta \theta + \operatorname{div} \mathbf{v} = 0, \\ &(\mathbf{u}, \mathbf{v}, \theta)|_{t=0} = (\mathbf{u}_0, \mathbf{v}_0, \theta_0), \end{aligned} \tag{1}$$

in $\mathbb{R}^3$, where the unknowns are the vector fields $\mathbf{u} = (u_1, u_2, u_3)$, $\mathbf{v} = (v_1, v_2, v_3)$ and the functions θ and p .

Here, $\mathbf{u}$ and $\mathbf{v}$ are the **barotropic mode** and the **first baroclinic mode** of the velocity while θ and p denote the temperature and the pressure.

Problem (1) takes the name of *Tropical Climate Model (TCM) with dampings*, where $\alpha \geq 1$ and $\beta \geq 1$.

When there are no damping terms

- System (1) is the *standard* TCM, derived in [5];
- when $\theta = 0$, and $\operatorname{div} \mathbf{v} = 0$, (1) is the *viscous Magneto Hydro Dynamics Equations*, where $\mathbf{v}$ stands for the *magnetic field*;
- when $\theta = \mathbf{v} = 0$, the system is the celeb viscous *Navier-Stokes Equations* for incompressible flow.

A few words on the model

The *standard* TCM, that is (1) without dampings, was introduced in the seminal work [5] in the context of *meteorology*, in order to **model large-scale atmosphere dynamics**.

Roughly speaking, TCM arises approximating the primitive equations with the ***Hydrostatic Boussinesq Equations on the Equatorial β -plane*** for the velocity field $\mathbb{U}$.

From a vertical decomposition of $\mathbb{U}$, they result defined the **barotropic mode** $\mathbf{u}$ and the baroclinic mode of $\mathbb{U}$. The **first baroclinic mode** $\mathbf{v}$ is nothing else the truncation up to the first term of a Galerkin expansion of the baroclinic mode.

Previous results on global well-posedness of solutions of TCM

In 2D no damping is needed.

- [7] proved the **global well-posedness** of (strong) solutions, when the system considered has viscosities only in the equations for $\mathbf{u}$ and $\mathbf{v}$ (but no for θ).

For the **3D** case, the question of global regularity for smooth solutions **remains generally open**.

- [6] considered 3D TCM with **fractional velocity diffusion** and provided **local well-posedness** of solutions.

For the case **3D with dampings**:

- [9] introduced TCM with damping **only** on $\mathbf{u}$. The **global existence and uniqueness** of strong solutions is established, **provided that $\alpha \geq 4$** .
- [10]: Damping $|\theta|^{\kappa-1} \theta$ also on the equation for the temperature. **Global regularity** if

(i) $\alpha, \beta \geq 4,$

(ii) $\frac{7}{2} \leq \alpha < 4, \quad \beta \geq \frac{5\alpha+7}{2\alpha}, \quad k \geq \frac{7}{2\alpha-5},$

(iii) $3 < \alpha \leq \frac{7}{2}, \quad \beta, k \geq \frac{7}{2\alpha-5}.$

Main result - Global regularity criterion when $3 \leq \alpha, \beta < 4$

Assumptions. We assume $3 \leq \alpha, \beta < 4$, and $\mathbf{u}_0, \mathbf{v}_0, \theta_0 \in H^s$, for some $3/2 < s \leq 2$, with $\operatorname{div} \mathbf{u}_0 = 0$. Here,

$$H^s = \text{Sobolev space } W^{s,2}.$$

Let $(\mathbf{u}, \mathbf{v}, \theta)$ be the local solution of (1), defined on some time interval $[0, T)$, with $0 < T < \infty$, and having the following regularity

$$\mathbf{u}, \mathbf{v}, \theta \in C([0, T]; H^s), \quad \text{and} \quad \mathbf{u}, \mathbf{v} \in L^2(0, T; H^{s+1}). \tag{2}$$

We obtain the following **global regularity criterion in terms of Homogeneous Besov spaces of $\mathbf{u}$ and $\mathbf{v}$** :

Theorem 1 (B.-Bisconti-Catania [1])

The local solution $(\mathbf{u}, \mathbf{v}, \theta)(t)$ can be extended beyond time T , with the same regularity as in (2), provided that

$$\int_0^T (\|\mathbf{u}(\tau)\|_{\dot{B}}^{\delta} + \|\mathbf{v}(\tau)\|_{\dot{B}}^{\gamma}) \, d\tau < \infty,$$

where

$$\delta = \frac{6(\alpha-1)}{3\alpha-5} > 2 \quad \text{and} \quad \gamma = \frac{6(\beta-1)}{3\beta-5} > 2,$$

and $\dot{B}$ is the Homogeneous Besov space $\dot{B}_{\infty, \infty}^0$.

Remark 1. From (2), by applying a regularity criterion in [8] we obtain the smoothness of global solutions.

Remark 2. It is highlighted the importance of the choice $\alpha, \beta \geq 4$ in order to obtain smooth solutions to (1), without additional constraints ([9, 10]).

Homogeneous Besov spaces. By means of a Littlewood-Paley dyadic decomposition of unity $\{\phi_j\}_{j \in \mathbb{Z}}$, for $s \in \mathbb{R}$, $1 < q \leq \infty$, we can define

$$\|\mathbf{f}\|_{\dot{B}_{\infty, q}^s} := \begin{cases} \left(\sum_{j \in \mathbb{Z}} 2^{js} \|\phi_j \star \mathbf{f}\|_{L^\infty}^q \right)^{1/q}, & \text{if } 1 < q < \infty, \\ \sup_{j \in \mathbb{Z}} \left\{ 2^{js} \|\phi_j \star \mathbf{f}\|_{L^\infty} \right\}, & \text{if } q = \infty. \end{cases}$$

The homogeneous Besov spaces $\dot{B}_{\infty, q}^s = \dot{B}_{\infty, q}^s(\mathbb{R}^3)$ are defined as

$$\dot{B}_{\infty, q}^s := \left\{ \mathbf{f} \in \mathcal{S}'(\mathbb{R}^3) / \mathcal{P}(\mathbb{R}^3) : \|\mathbf{f}\|_{\dot{B}_{\infty, q}^s} < \infty \right\}$$

where $\mathcal{S}'$ is the space of tempered distributions and $\mathcal{P}$ is the space of polynomials.

Key estimates that bring $\dot{B}_{\infty, q}^0$ into play

Let BMO be the space of functions of *bounded mean oscillation* in $\mathbb{R}^3$; Together with the fact that

- $\dot{B}_{\infty, 2}^0 \subset \text{BMO}$,
- $\text{BMO} \subset \dot{B}_{\infty, \infty}^0$,

in order to estimate the BMO-norm **by means of s-logarithmic growths**, we use the following formula [4]

$$\|\mathbf{f}\|_{\dot{B}_{\infty, 2}^0} \lesssim 1 + \|\mathbf{f}\|_{\dot{B}_{\infty, \infty}^0} \ln^{\frac{1}{2}}(e + \|\mathbf{f}\|_{H^s}), \quad s > 3/2, \tag{3}$$

$\|\mathbf{f}\|_X \lesssim \|\mathbf{f}\|_Y$ means $\|\mathbf{f}\|_X \leq C \|\mathbf{f}\|_Y$ with C independent of $\mathbf{f}$.

- Estimate (3) is used together with the following **BMO-estimates of L^p -products** [3], $p > 1$,

$$\|\mathbf{f} \mathbf{g}\|_{L^p} \lesssim \|\mathbf{f}\|_{L^p} \|\mathbf{g}\|_{\text{BMO}} + \|\mathbf{f}\|_{\text{BMO}} \|\mathbf{g}\|_{L^p}. \tag{4}$$

Scheme of the proof of Theorem 1

Let $\|\cdot\| := \|\cdot\|_{L^2}$. We show explicitly:

$$\sup_{\tau \in (0, T)} \|\Lambda^s(\mathbf{u}, \mathbf{v}, \theta)(\tau)\|^2 < \infty, \tag{5}$$

where $\Lambda^s = (-\Delta)^{\frac{s}{2}}$, i.e. $\widehat{(-\Delta)^s f}(\xi) = |\xi|^s \widehat{f}(\xi)$, for $\xi \in \mathbb{R}^n$, in terms of Fourier transform.

The proof consists in proving **suitable energy estimates**, by properly taking advantage of (3) and (4), as well as the fractional Gagliardo-Nirenberg's and Young's inequalities.

H^1 -estimates. By taking the L^2 -inner product of $(1)_1$, $(1)_2$ and $(1)_3$ with $-\Delta \mathbf{u}$, $-\Delta \mathbf{v}$ and $-\Delta \theta$, resp., integrating by parts and using **(4)-(3)**, Gagliardo-Nirenberg's and Young's inequalities, one obtains

$$\begin{aligned} &\frac{d}{dt} (\|\nabla(\mathbf{u}, \mathbf{v}, \theta)(t)\|^2) + \text{positive term} \\ &\lesssim \|\nabla(\mathbf{u}, \mathbf{v}, \theta)\|^2 \left(\mathcal{A} + (\mathcal{A} + \|\mathbf{v}\|_{\dot{B}}^{\gamma} + \|\mathbf{u}\|_{\dot{B}}^{\delta}) \ln(e + \|(\mathbf{u}, \mathbf{v})\|_{H^s}) \right), \end{aligned}$$

where for $\mathcal{A} = \|\mathbf{u}\|_{\alpha+1}^{\alpha+1} + \|\mathbf{v}\|_{\beta+1}^{\beta+1}$ it holds

$$\int_0^T \mathcal{A}(\tau) \, d\tau \leq \|(\mathbf{u}_0, \mathbf{v}_0, \theta_0)\|^2,$$

as a consequence of the L^2 -inner product of (1) with $(\mathbf{u}, \mathbf{v}, \theta)$.

- For every $\varepsilon > 0$ there exists $0 < t_* < t$ such that, by integrating on $[t_*, t]$ the above differential inequality by means of the **Gronwall's lemma** for differential inequalities, one deduces

$$\begin{aligned} &\|\nabla(\mathbf{u}, \mathbf{v}, \theta)(t)\|^2 + \int_{t_*}^t \text{positive terms} \\ &\leq \|\nabla(\mathbf{u}, \mathbf{v}, \theta)(t_*)\|^2 e^{C \int_{t_*}^t (1 + \|\mathbf{u}\|_{\dot{B}}^{\delta} + \|\mathbf{v}\|_{\dot{B}}^{\gamma}) \ln(e + \mathbf{y}(\tau)) \, d\tau} \\ &\leq C_* (e + \mathbf{y}(t))^{C\varepsilon}, \end{aligned}$$

where we set

$$\mathbf{y}(t) := \sup_{[t_*, t]} \|(\mathbf{u}, \mathbf{v}, \theta)\|_{H^s}, \quad \text{and} \quad C_* \approx \|\nabla(\mathbf{u}, \mathbf{v}, \theta)(t_*)\|.$$

The value of $0 < t_* < t$ is chosen to satisfy

$$\int_{t_*}^t (\|\mathbf{u}(\tau)\|_{\dot{B}}^{\delta} + \|\mathbf{v}(\tau)\|_{\dot{B}}^{\gamma}) \, d\tau < \varepsilon.$$

H^s -estimates, $3/2 < s \leq 2$. Apply Λ^s to (1) and multiply in L^2 the resulting equations, by $(\Lambda^s \mathbf{u}, \Lambda^s \mathbf{v}, \Lambda^s \theta)$. This, with the help of Gagliardo-Nirenberg's, Young's, and (in particular) **Kato-Ponce's inequality**, leads to

$$\frac{d}{dt} (\|\Lambda^s(\mathbf{u}, \mathbf{v}, \theta)(t)\|^2) + \|\Lambda^{s+1}(\mathbf{u}, \mathbf{v}, \theta)(t)\|^2 \leq \mathfrak{C}(t), \tag{6}$$

where, for some $C, c_i, \rho_i \geq 0$,

$$\mathfrak{C}(t) := C + c_1 \|\nabla \mathbf{u}(t)\|^{\rho_1} + c_2 \|\nabla \mathbf{v}(t)\|^{\rho_2} + c_3 \|\nabla \theta(t)\|^{\rho_3}.$$

- Similarly as observed above, for any $\varepsilon > 0$ there exists $0 \leq t_* < t < T$ such that

$$\int_{t_*}^t \mathfrak{C}(\tau) \, d\tau \leq C_*(t - t_*)(e + \mathbf{y}(t))^{C\varepsilon}. \tag{7}$$

Formula (5) follows from (6), (7) and Gronwall's lemma.

A second regularized TCM: 3D Bardina Tropical Climate Model

Consider the following problem, in the 3D torus $\mathbb{T}^3$,
 $(I - \alpha^2 \Delta) \partial_t \mathbf{w} - (I - \alpha^2 \Delta) \Delta \mathbf{w} + (\mathbf{w} \cdot \nabla) \mathbf{w} + \operatorname{div}(\mathbf{b} \otimes \mathbf{b}) + \nabla p = \mathbf{f},$
 $(I - \alpha^2 \Delta) \partial_t \mathbf{b} - (I - \alpha^2 \Delta) \Delta \mathbf{b} + (\mathbf{w} \cdot \nabla) \mathbf{b} + (\mathbf{b} \cdot \nabla) \mathbf{w} = \nabla \theta,$
 $\partial_t \theta + (\mathbf{w} \cdot \nabla) \theta - \Delta \theta + \operatorname{div} \mathbf{b} = 0,$
 $\operatorname{div} \mathbf{w} = 0,$
 $(\mathbf{w}, \mathbf{b}, \theta)|_{t=0} = (\bar{\mathbf{u}}_0, \bar{\mathbf{v}}_0, \theta_0).$

The above system is formally obtained by the standard TCM by applying the filter $G_\alpha := (I - \alpha^2 \Delta)^{-1}$ to the equations for $\mathbf{u}$ and $\mathbf{v}$: $\mathbf{w} = \mathbf{u} := G_\alpha \mathbf{u}$, $\mathbf{b} = \mathbf{v} := G_\alpha \mathbf{v}$.

The main result - Existence of a global attractor

Let $\mathbf{f} \in L^2(\mathbb{T}^3)$, $\mathbf{u}_0, \mathbf{v}_0 \in L^2(\mathbb{T}^3)$, with $\operatorname{div} \mathbf{u}_0 = 0$, and $\theta \in H^1(\mathbb{T}^3)$. There exists a unique global solution $(\mathbf{w}, \mathbf{b}, \theta)$. Furthermore, we obtain the following **existence of a global attractor**:

Theorem 2 (B.-Bisconti-Catania [2])

There exists $\mathcal{A} \subset H_o^1(\mathbb{T}^3) \times H^1(\mathbb{T}^3) \times L^2(\mathbb{T}^3)$, unique compact global attractor in terms of the solution $(\mathbf{w}, \mathbf{b}, \theta)$, where the subscript “ σ ” denotes divergence-free functions.

The result contained in Theorem 2 follows essentially from standard *theory of absorbing sets* once that decay estimates for higher order energy is proved. More precisely, the crucial inequality is the following

$$\begin{aligned} &\limsup_{t \rightarrow \infty} \|\nabla(\mathbf{w}, \mathbf{b}, \theta)(t)\|^2 + \alpha^2 \|\Delta(\mathbf{w}, \mathbf{b})(t)\|^2 \leq \\ &\Phi \left(\|\mathbf{f}\|, \|\Lambda^{-1} \mathbf{f}\|, \|\Lambda^{-2} \mathbf{f}\|, \alpha, \lambda_1 \right), \end{aligned}$$

where Φ is a rational function, λ_1 is the first eigenvalue of the Stokes operator on $\mathbb{T}^3$ and $\Lambda^{-1} \mathbf{f}$, $\Lambda^{-2} \mathbf{f}$ are defined as done for positive orders, in terms of Fourier transform.

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